MIL-C-6783

23 June 1950

MILITARY SPECIFICATION

CLAMPS; BATTERY HOLD-DOWN

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

AN-C-144
30 April 1945

This specification consists of this cover sheet and Specification AN-C-144, dated 30 April 1945, with Amendment -1, dated 17 July 1945, without modification.

Copies of this specification may be obtained from the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio; or the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.

When a request for this specification is received by a supplying activity, it will be necessary to attach this cover sheet to the pertinent specification before issue.

Custodian:
Navy - BuAer

Other interest:
Air Force

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AN-C-144
30 April 1945

ARMY-NAVY AERONAUTICAL SPECIFICATION
CLAMPS; BATTERY HOLD-DOWN

This specification was approved on the above date by joint action of the War and Navy Departments, for use in the procurement of aeronautical supplies and shall become effective not later than 30 October 1945. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS.

A-1. The following specifications and drawings of the issue in effect on date of invitation for bids shall form a part of this specification:

A-1a. AN Aeronautical Specifications.-

AN-GGG-S-126	Screw-Threads; Standard, Aircraft
AN-P-32	Plating; Zinc
AN-P-61	Plating; Cadmium
AN-QQ-S-91	Salt-Spray-Corrosion Test; Process for
AN-QQ-A-696	Anodic Films; Corrosion-Protective (for) Aluminum Alloys
AN-B-17	Boxes; General Specification for Wood-Cleated Solid Fiberboard (For Export Shipment)

A-1b. Other Government Specification.-

100-14:39P16	Packaging and Packing for Overseas Shipment; Army-Navy General Specification for.

A-1c. AN Aeronautical Standard Drawing.-

AN3156	Clamp-Battery Hold-Down.

A-2. Special requirements of the individual Departments of the Government are noted under section H.

B. TYPE.

B-1. This specification covers one type of battery hold-down clamp.

C. MATERIAL AND WORKMANSHIP.

C-1. Materials.- Materials used in the manufacture of the hold-down clamps shall be of high quality and suitable for the purpose intended.

C-2. Workmanship.- Workmanship shall be in accordance with the best practice for this type of equipment.

D. GENERAL REQUIREMENTS.

D-1. See section E.

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E. DETAIL REQUIREMENTS.

E-1. Design and Construction.- The battery hold-down clamp shall be so designed and constructed that no parts will work loose in service and shall be in accordance with Drawing AN3156. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

E-1a. Material.- Dependent on design, the material for the clamp may be steel or aluminum or any other material which meets the requirements of this specification and acceptable to the Procuring Agency.

E-1b. Operation.- The operational characteristics of the clamp shall be such that when in the open position, the clamp shall allow enough clearance so that it may easily be taken off the battery hold-down strap to allow removal of the battery; when in the locked position, the clamp shall render sufficient tension to hold the battery securely in place and shall have a positive locking action.

E-1c. Cam.- The cam action necessary to accomplish the desired operation is specified on Drawing AN3156.

E-1d. Weight.- The weight of the battery hold-down clamp shall not exceed 3 ounces.

E-2. Screw Threads.- Screw threads shall be in accordance with Specification AN-GGG-S-126.

E-3. Finish.- When materials are used in the construction of the clamp which are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

E-3a. Aluminum and aluminum alloys shall be anodized in accordance with Specification AN-QQ-A-696.

E-3b. Cadmium plating shall be in accordance with Specification AN-P-61.

E-3c. Zinc plating shall be in accordance with AN-P-32.

E-4. Marking.- The battery hold-down clamp shall be legibly and permanently marked with the AN part number, the manufacturer's part number, and the manufacturer's name or trademark.

E-5. Performance.- The battery hold-down clamp shall satisfy the performance requirements as specified in section F under the paragraphs headed as follows:

(a) Vibration
(b) Corrosion Resistance.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F-1. Inspection.- When inspection is conducted at the contractor's plant, the tests shall be accomplished by the contractor under the supervision of the Inspector.

F-2. Laboratory Facilities.- Contractors not having laboratory facilities satisfactory to the Inspector shall engage the services of a commercial testing laboratory acceptable to the Inspector.

F-3. Acceptance.- Previous acceptance or approval of material or release of any design by the Government shall in no case be construed as a guaranty of the acceptance of the finished product.

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F-4. Reports of Tests.- The contractor shall maintain a record, available to the Inspector, of the quantitative results of all inspection tests required by this specification.

F-5. Examination of Product.- Each battery hold-down clamp shall be carefully examined to determine conformance with this specification with respect to Material and Workmanship, Dimensions, Weight, and Finish.

F-6. Inspection Tests.- Five battery hold-down clamps selected at random by the Inspector from each lot of 2500 or less, or direct ratio thereto dependent upon the total quantity of clamps under contract or order, shall be subjected to the following tests. A lot shall consist of all the clamps of the same part number which are submitted for acceptance under the same contract or order.

F-6a. Vibration.- The clamp shall be installed on a mock assembly and then vibrated for a period of three hours on a suitably designed vibration stand. The amplitude of vibration shall be 0.03 inch (maximum total excursion of 0.06 inch) and a frequency varied uniformly between the approximate limits of 10 to 55 cycles per second. The clamp shall not become loosened in any manner as a result of the vibration.

F-6b. Corrosion Resistance.- The clamp shall be subjected for six hours to a salt spray test in accordance with Specification AN-QQ-S-91. Upon completion of the test, the clamp shall be installed in a mock assembly and tested for ease of operation. There shall be no sticking between moving parts and the operation of the clamp shall be smooth and easy.

F-7. Rejection and Retest.- When any representative sample fails to meet the requirements of the representative sample tests, five additional clamps shall be selected from the lot represented and shall be subjected to the test at which the representative sample failed. If any of these samples fail to meet the test, the lot represented shall be rejected. Clamps failing to meet the test requirements shall be rejected and returned at the contractor's expense. Clamps which have been rejected may be reworked or replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects shall be furnished the Inspector. Clamps rejected after retest shall not be resubmitted without the specific approval of the Procuring Agency.

G. PACKAGING, PACKING AND MARKING FOR SHIPMENT.

G-1. General.- Packaging, packing, and marking requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

G-2. Interior Packaging.- Unless otherwise specified, each battery hold-down clamp shall be wrapped or bagged in a suitable wrapping. Twelve clamps shall then be packaged within an interior box in accordance with Specification 100-14:39P16, section headed "Folding and Set-up Boxes".

G-3. Exterior Packing.- Unless otherwise specified, all items shall be packed for export shipment. Shipping containers, insofar as possible, shall contain identical number of articles, shall be of a uniform size and shall be designed to enclose the contents in a snug, tight fitting manner. The gross weight of the exterior shipping container when packed for shipment shall not exceed approximately 200 pounds.

G-3a. Domestic Shipment.- Unless otherwise specified, for domestic shipment interior packages shall be packed in substantial commercial exterior shipping containers so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of Consolidated Freight Classification Rules in effect at the time of shipment. The use of corrugated or solid fiberboard having a minimum Mullen test of less than 200 pounds is prohibited. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

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G-3b. Export Shipment.- Unless otherwise specified, for export shipment, the interior packages shall be packed in exterior shipping containers in accordance with Specification 100-14:39P16, sections headed "Cleated Plywood Shipping Boxes", "Nailed Wood Shipping Boxes", or boxes in accordance with Specification AN-B-17.

G-4. Marking.-

G-4a. Interior Packages.- Each interior package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened:

CLAMP - BATTERY HOLD-DOWN
AN3156 - (proper dash number)
Manufacturer's part number _____
Name of contractor (and name of manufacturer if not the same)
Government order number (or contract number if order number is not assigned)

G-4b. Exterior Shipping Container.- Each exterior shipping container shall be marked as specified in section H.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H-1. The following departmental specifications of the issue in effect on date of invitation for bids shall form a part of this specification applicable to purchases by the agency indicated.

H-1a. Army.- U. S. Army Specification 94-40645, Marking; Shipment (Domestic and Export). Copies of U. S. Army specifications may be obtained from the Army Air Forces activities listed in section I.

H-1b. Navy.- Navy Shipment Marking Handbook. Copies of this handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.

I. NOTES.

I-1. Use.- The battery hold-down clamp is used, as shown on Drawing AND10441, to hold the battery in place.

I-2. Ordering Data.- Requisitions, contracts, and orders should state the AN part number of the clamp desired.

I-3. Specifications.- When requesting specifications, refer to both title and number.

I-3a. Sources.- Copies of Army-Navy Aeronautical Specifications, and Drawings, and Specification 100-14:39P16, may be obtained upon application to the Commanding General in the following Air Technical Service Command Districts: New York, Atlanta, Detroit, Chicago, Wichita, and Los Angeles; to the Director, Army Air Forces, Headquarters, Air Technical Service Command, Wright Field, Dayton, Ohio; or to the Bureau of Aeronautics, Navy Department, Washington 25, D. C. Naval activities should make application to the Commanding Officer, Naval Aircraft Modification Unit, Johnsville, Pa.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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